Liner Xiang

Department of Statistics, Donald Bren School of Information and Computer Sciences, University of California
Irvine, Bren Hall 2019, Irvine, CA 92697-1250
Phone: 9492328227 Email: linerx1@uci.edu Homepage: https://linerxiang.github.io/

EDUCATION

University of California, Irvine September 2022-June 2027(expected)
School: Department of Statistics, Donald Bren School of Information and Computer Sciences
Degree Expected: Ph.D. in Statistics
Co-advisors: Hengrui Cai and Weining Shen
GPA: 3.99/4.0

University of Science and Technology of China September 2018-June 2022
School: School of the Gifted Young
Degree: B.S. in Statistics

AWARDS

NeurIPS 2026 Financial Assistance Award October 2026
Outstanding Graduate Teaching Assistant Award, Department of Statistics, UC Irvine September 2025
MBZUAI AI Forum Travel Award, MBZUAI May 2025
Excellent Student Scholarship, USTC September 2018, 2019, 2020, 2021

RESEARCH INTERESTS

My research lies at the intersection of **LLM alignment and evaluation**, **LLM reasoning**, **online decision-making**, and **causal inference**. I develop principled statistical and optimization frameworks for evaluating and improving large language models under weak supervision and distribution shift, with an emphasis on off-policy evaluation, robust routing, and cost-efficient LLM-as-a-judge systems. More broadly, I study sequential decision-making under interference and strategic interaction, combining reinforcement learning, game theory, and doubly robust methods to provide theoretical guarantees alongside practical, scalable algorithms.

PUBLICATIONS & PREPRINTS

* Equal contribution

Xiang, L., Jiang, H., Liu, Y., Jiang, J., Cai, H., and Song, R. (2026).
Counterfactual Outcome Prediction in Foundation Models: An Inverse Probability Weighting Approach. ***(X-IPW)***
Under review at ACL Rolling Review (ARR), 2026.

Xiang, L., Zhang, W., and Cai, H. (2026).
OTROPE: Optimal Transport-based Robust Off-policy Evaluation for Large Language Models. ***(OTROPE)***
Advances in Neural Information Processing Systems (NeurIPS), 2026.

Zhuang, X.**, Xiang, L.,** and Zhou, W. (2026).
When and How to Collaborate: State-Adaptive Sequential Multi-Agent Collaboration via Reinforcement Learning. ***(RESC)***
Under review at ACL Rolling Review (ARR), 2026.

Zhang, W.***,** Zhang, L.***, Xiang, L.****, and Cai, H. (2026).
Reasoning Is Not Free: Robust Adaptive Cost-Efficient Router for LLM-as-a-Judge. ***(RACER)***
International Conference on Machine Learning (ICML), 2026.

Zhang, L.***, Xiang, L.****, Zhang, W.***,** Zheng, Y., and Cai, H. (2026).
Large Language Model Alignment with Complex Feedback: A Survey. ***(Survey)***
Under review at ACL Rolling Review (ARR), 2026.

Xiang, L., Wang, J., and Cai, H. (2025).
Foresighted Online Policy Optimization with Interference. ***(FRONT)***
Invited revision at Journal of Machine Learning Research (JMLR).

Xiang, L., Wang, Y., and Cai, H. (2025).
Policy Optimization and Statistical Inference for Online Contextual Matrix Games. ***(OnGameLearn)***
Revised and resubmitted after major revision, Journal of the American Statistical Association (JASA).

CROSS-DISCIPLINARY RESEARCH

Alantari, H., **Xiang, L.**, Cai, H., Shen, W., and Currim, I.
Modeling Book Lovers' Sequential Decisions to Read, Rate, and Review: Missing Data Imputation for Efficient Estimation.
Revised and resubmitted after major revision, European Journal of Marketing.

Ding, X.*, Meng, Y.*, **Xiang, L.***, and Boden-Albala, B. (2024).
Stroke recurrence prediction using machine learning and segmented neural network risk factor aggregation.
Discover Public Health, 21(1).

INTERNSHIP EXPERIENCES

Applied Scientist | Amazon Seattle, WA
Mentors : Yang Liu and Rui Song June 2026-September 2026

- Developed a causal fine-tuning framework **X-IPW** for pretrained sequence models to improve counterfactual outcome prediction under confounded sequential treatment assignment.
- Designed an inverse-propensity-weighted training objective with joint propensity and outcome modeling to correct behavior-policy selection bias and improve learning for underrepresented treatment–context combinations.
- Evaluated the model-agnostic framework on real-world data across multiple sequence-modeling backbones, demonstrating improved counterfactual prediction performance over standard fine-tuning.

Statistician | Quantum Leap Healthcare Collaborative San Francisco, CA
Mentors: Peter Norwood and Philip Beineke June 2025-September 2025

- Contributed to the design of the I-SPY2 adaptive breast cancer trial using Bayesian modeling and reinforcement learning.
- Built an interactive trial-planning tool to help clinicians compare adaptive treatment strategies before deployment.

RESEARCH EXPERIENCES

When and How to Collaborate: State-Adaptive Sequential Multi-Agent Collaboration via Reinforcement Learning Irvine, CA
Advisor: Wenzhuo Zhou March 2026-August 2026

- Developed **RESC**, a reinforcement learning framework for state-adaptive sequential multi-agent collaboration.
- Designed a trajectory-wise DPO-based routing policy to dynamically select agents from a heterogeneous agent pool.
- Evaluated RESC on question answering and open-ended report generation, showing improved performance.

OTROPE: Optimal Transport-based Robust Off-policy Evaluation for Large Language Model Irvine, CA
Advisor: Hengrui Cai October 2025-April 2026

- Formulated model-shifted LLM evaluation as a semi-supervised off-policy evaluation problem under limited human annotation and behavior–target distribution shift.
- Developed **OTROPE**, an optimal transport-based reweighting method in semantic embedding space that avoids token-level likelihoods, behavior-policy modeling, and sequence-level density-ratio estimation while incorporating doubly robust-style correction.
- Proved consistency and convergence guarantees for OTROPE, characterized failure modes of standard PPI/DR estimators under LLM distribution shift, and demonstrated strong empirical performance on synthetic and real LLM evaluation tasks.

Reasoning Is Not Free: Robust Adaptive Cost-Efficient Routing for LLM-as-a-Judge Irvine, CA
Advisor: Hengrui Cai September 2025-January 2026

- Conducted controlled studies showing that reasoning LLM judges significantly improve evaluation accuracy on structured tasks (e.g., math, coding) but incur high computational cost and limited gains on simpler tasks.
- Formulated judge selection as a constrained distributionally robust optimization problem under distribution shift and proposed **RACER**, a routing framework that adaptively switches between reasoning and non-reasoning judges under a fixed budget.
- Developed an efficient primal–dual algorithm with theoretical guarantees (uniqueness and linear convergence) and demonstrated strong accuracy–cost trade-offs through extensive experiments.

Foresighted Online Policy Optimization with Interference Irvine, CA
Advisor: Hengrui Cai and Jiayi Wang March 2024-December 2024

- Developed a long-term policy framework for sequential decision-making under cross-user interference, formulating the optimal objective and establishing regret guarantees.

- Validated the framework through simulations and synthetic experiments in contextual bandits, demonstrating strong long-term empirical performance over existing myopic approaches.

Policy Optimization and Statistical Inference for Online Contextual Matrix Games Irvine, CA
Advisor: Hengrui Cai and Yixin Wang January 2024-August 2025

- Developed an optimal policy learning framework for two-player zero-sum games, integrating game-theoretic structure with doubly robust evaluation and establishing convergence guarantees.
- Validated the approach through simulations on matrix games and real data, demonstrating strong empirical performance and reliable policy evaluation.

ACADEMIC SERVICE

Reviewer: ICLR (2025, 2026, 2027), ICML (2026), NeurIPS (2026), Biostatistics, Annals of Applied Statistics.

PRESENTATIONS

Internship Presentation Amazon, 2026
Causal Fine-Tuning for Foundation Models: Leveraging Causal Structure in Sequential Data for Outcome Simulator

Research Presentation UC Irvine, 2025
Reliable Evaluation of LLMs through Rectified Estimation

Advancement to Candidacy Presentation UC Irvine, 2025
Decisions in the Wild: Foresight, Adversaries, and Human Feedback in Online Policy Optimization and Evaluation